Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.12_k-1)

Metric

$$ds^2 = -P\left(r\right)\mathrm{d}t\otimes\mathrm{d}t + \frac{1}{P\left(r\right)}\mathrm{d}r\otimes\mathrm{d}r + r^2\mathrm{d}\theta\otimes\mathrm{d}\theta + r^2\sinh\left(\theta\right)^2\mathrm{d}\phi\otimes\mathrm{d}\phi$$

Coordinates

$$\left(x^\mu\right) = \left(t, r, \theta, \phi\right)$$

Metric Functions

$$P = -1 - \frac{1}{3}\,\Lambda r^2 + \frac{e^2}{r^2} - \frac{2\,m}{r}$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\,\sqrt{2}\sqrt{P\left(r\right)}\right)\,dt + \left(\frac{\sqrt{2}}{2\,\sqrt{P\left(r\right)}}\right)\,dr$$

$$n = \left(\frac{1}{2}\,\sqrt{2}\sqrt{P\left(r\right)}\right)\,dt + \left(-\frac{\sqrt{2}}{2\,\sqrt{P\left(r\right)}}\right)\,dr$$

$$m = \left(\frac{1}{2}\,\sqrt{2}r\right)\,d\theta + \left(\frac{1}{2}i\,\sqrt{2}r\sinh\left(\theta\right)\right)\,d\phi$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}r\right)\,d\theta + \left(-\frac{1}{2}i\,\sqrt{2}r\sinh\left(\theta\right)\right)\,d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\,\sqrt{P\left(r\right)}}\right)\,dt + \left(\frac{1}{2}\,\sqrt{2}\sqrt{P\left(r\right)}\right)\,dr$$

$$n = \left(-\frac{\sqrt{2}}{2\,\sqrt{P\left(r\right)}}\right)\,dt + \left(-\frac{1}{2}\,\sqrt{2}\sqrt{P\left(r\right)}\right)\,dr$$

$$m = \left(\frac{\sqrt{2}}{2\,r}\right)\,d\theta + \left(\frac{i\,\sqrt{2}}{2\,r\sinh\left(\theta\right)}\right)\,d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,r}\right)\,d\theta + \left(-\frac{i\,\sqrt{2}}{2\,r\sinh\left(\theta\right)}\right)\,d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\,\sqrt{P\left(r\right)}}\,\partial_t + \frac{1}{2}\,\sqrt{2}\sqrt{P\left(r\right)}\,\partial_r$$

$$\Delta = -\frac{\sqrt{2}}{2\,\sqrt{P\left(r\right)}}\,\partial_t - \frac{1}{2}\,\sqrt{2}\sqrt{P\left(r\right)}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2\,r}\,\partial_\theta + \frac{i\,\sqrt{2}}{2\,r\sinh\left(\theta\right)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,r}\,\partial_\theta - \frac{i\,\sqrt{2}}{2\,r\sinh\left(\theta\right)}\,\partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}\sqrt{P\left(r\right)}}{2\,r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\sqrt{P\left(r\right)}}{2\,r}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cosh\left(\theta\right)}{4\,r\sinh\left(\theta\right)}$$

$$\beta = \frac{\sqrt{2}\cosh\left(\theta\right)}{4\,r\sinh\left(\theta\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}\frac{\partial}{\partial r}P\left(r\right)}{8\,\sqrt{P\left(r\right)}}$$

$$\gamma = \frac{\sqrt{2}\frac{\partial}{\partial r}P\left(r\right)}{8\,\sqrt{P\left(r\right)}}$$

Ricci Tensor Components

$\Phi_{00} = 0$

$\Phi_{01} = 0$

$\Phi_{02} = 0$

$\Phi_{10} = 0$

$$\Phi_{11} = -\frac{P\left(r\right)}{4\,r^{2}} - \frac{1}{4\,r^{2}} + \frac{1}{8}\,\frac{\partial^{2}}{(\partial r)^{2}}P\left(r\right)$$

$\Phi_{12} = 0$

$\Phi_{20} = 0$

$\Phi_{21} = 0$

$\Phi_{22} = 0$

$$\Lambda = -\frac{\frac{\partial}{\partial r}P\left(r\right)}{6\,r} - \frac{P\left(r\right)}{12\,r^{2}} - \frac{1}{12\,r^{2}} - \frac{1}{24}\,\frac{\partial^{2}}{(\partial r)^{2}}P\left(r\right)$$

Weyl Tensor Components

$\Psi_{0} = 0$

$\Psi_{1} = 0$

$$\Psi_2 = -\frac{\frac{\partial}{\partial r} P\left(r\right)}{6\,r} + \frac{P\left(r\right)}{6\,r^2} + \frac{1}{6\,r^2} + \frac{1}{12}\,\frac{\partial^2}{(\partial r)^2} P\left(r\right)$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"